Day 5: Troubleshooting Network Services Practical (1 hour)

1.Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Aim: To simulate a DNS resolution failure by mapping www.spotify.com to an incorrect IP address and observing the ping result.

Procedure (Windows)

1. Open Notepad as Administrator.
2. Open the hosts file from this path:

C:\Windows\System32\drivers\etc\hosts

In the Open dialog, select All Files (.) to view the hosts file.

3. Add the following line at the end of the file:

192.0.2.1 www.spotify.com

192.0.2.1 is a reserved documentation IP address used here as an incorrect destination.

4. Save the file.
5. Open Command Prompt (CMD) and run:

ipconfig /flushdns

6. Now ping Spotify:

ping www.spotify.com

Expected Output

The output may look similar to this:

Pinging www.spotify.com [192.0.2.1] with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.0.2.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Observation

The domain www.spotify.com resolves to the incorrect IP address 192.0.2.1 because of the hosts-file entry. The ping requests fail to receive replies, resulting in packet loss.

2.Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Aim: To configure a DHCP server in Cisco Packet Tracer with an incorrect default gateway, verify the configuration received by a client, and troubleshoot the issue.

Procedure

1. Open Cisco Packet Tracer.
2. Add one Server, one Switch, and one PC to the workspace.
3. Connect the server and PC to the switch using Copper Straight-Through cables.
4. Configure the server's static IP address:
    - IP Address: 192.168.1.2
    - Subnet Mask: 255.255.255.0
5. Open Server → Services → DHCP and turn DHCP On.
6. Create the following DHCP pool with an intentionally incorrect default gateway.

Setting	**Value**
Pool Name	Office_LAN
Default Gateway	192.168.1.254 (incorrect)
DNS Server	8.8.8.8
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	20

7. Click Add or Save, depending on the Packet Tracer version.
8. Click the PC → Desktop → IP Configuration → DHCP.
9. Open the PC's Command Prompt and enter:

ipconfig

Expected Observation

The PC should receive an IP address from the DHCP pool, along with the incorrect default gateway.

IP Address.....: 192.168.1.10

Subnet Mask.....: 255.255.255.0

Default Gateway...: 192.168.1.254

These are example values; your actual output may differ.

Troubleshooting and Fix

1. Identify the problem: Run ipconfig on the PC and check the default gateway.
2. Check the network configuration: Verify the correct gateway address for the LAN. For example, if a router uses 192.168.1.1, that should be the gateway.
3. Correct the DHCP pool: Open Server → Services → DHCP and change the Default Gateway from 192.168.1.254 to 192.168.1.1.
4. Renew the client's configuration: Select DHCP again on the PC or use the following commands if supported:
5. ipconfig /release

ipconfig /renew

6. Verify the fix: Run ipconfig again and check that the default gateway is now 192.168.1.1. If a router is connected, test it using:

ping 192.168.1.1

3.Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Aim: To diagnose and fix a DNS problem where devices can access websites using IP addresses but cannot access them using domain names.

Scenario

In a simulated network, a PC can connect to a website using its IP address, but cannot open it using its domain name, such as www.google.com. This indicates a possible DNS configuration or resolution problem.

Troubleshooting Steps

Step 1: Check network connectivity

Open Command Prompt and run:

ping 8.8.8.8

Observation: If replies are received, IP connectivity is working, although this alone does not prove that all network services are functioning correctly.

Step 2: Test DNS resolution

Run:

nslookup www.google.com

Example of a possible failure:

DNS request timed out.

Server: Unknown

Address: 192.168.1.254

DNS request timed out.

Observation: The configured DNS server is not responding to the query.

Step 3: Check the DNS server configuration

Run:

ipconfig /all

Check the DNS Servers entry. Verify that the configured DNS server address is correct and reachable.

Step 4: Test an alternative DNS server

Run:

nslookup www.google.com 8.8.8.8

If this query succeeds, the problem may be with the originally configured DNS server or the network path to it. The test may fail if external DNS queries are blocked.

Step 5: Fix the DNS settings

- Configure the client with the correct DNS server address.
- In Cisco Packet Tracer, open the DHCP server settings and correct the DNS Server field in the DHCP pool.
- Ensure the DNS server service is running and that DNS traffic (port 53, UDP/TCP as appropriate) is allowed.
- Renew the client's DHCP configuration if the DNS address is provided through DHCP.

Step 6: Clear the DNS cache and verify

Run:

ipconfig /flushdns

nslookup www.google.com

ping www.google.com

If the domain resolves to an IP address and the ping test succeeds, name resolution is working. Note that ping replies may still be blocked even when DNS works correctly.

Observation Table

Test	**Expected observation**
ping 8.8.8.8	Replies received if IP connectivity works
nslookup www.google.com	Timeout or DNS error before the fix
nslookup www.google.com 8.8.8.8	May succeed if the alternative DNS server is reachable
ipconfig /all	Shows the configured DNS server
nslookup www.google.com after the fix	Returns a domain name and IP address

**4.A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.**

Aim: To identify the reasons a device receives an APIPA address (169.254.x.x) instead of an IP address from the DHCP server and resolve the problem.

Possible Causes

1. Network connectivity issue: The Ethernet cable is disconnected, or the switch port is down.
2. DHCP server unavailable: The DHCP service is switched off or the server is powered down.
3. DHCP scope exhaustion: All IP addresses in the DHCP pool have been assigned to other devices.
4. Incorrect DHCP configuration: The IP range, subnet mask, or DHCP pool settings are incorrect.
5. DHCP relay issue: A router's DHCP relay configuration is missing or incorrect when the client and server are on different subnets.
6. Firewall blocking DHCP: DHCP traffic is blocked by a firewall or network access control.
7. VLAN mismatch: The client is connected to a different VLAN from the DHCP server, without a working DHCP relay.

Step-by-Step Troubleshooting Process

Step 1: Check the client's IP configuration

Open Command Prompt and execute:

ipconfig /all

Observation: If the device has an address such as 169.254.10.20, it likely failed to obtain an IPv4 address from DHCP and assigned itself an APIPA address.

Step 2: Check physical connectivity

- Verify that the Ethernet cable is connected properly.
- Check the switch port and link lights.

- In Cisco Packet Tracer, confirm that the connection links are green and the devices are powered on.

Step 3: Test DHCP renewal

Run:

ipconfig /release

ipconfig /renew

If the device still receives an APIPA address, continue troubleshooting.

Step 4: Verify DHCP server availability

- Check whether the DHCP server is powered on and connected to the network.
- Confirm that the DHCP service is enabled.
- In Packet Tracer, open Server → Services → DHCP and ensure the service is set to On.

Step 5: Check DHCP scope availability

Verify the configured start IP address, subnet mask, and maximum number of users.

For example, if the DHCP pool contains only five usable addresses and all five are leased, new clients may fail to obtain an address. Expand the pool or release unused leases as appropriate.

Step 6: Verify network settings

- Confirm that the client is configured to obtain an IP address automatically.
- Check that the DHCP server's pool matches the client's subnet.
- If the client is on another subnet, verify that the router's DHCP relay (ip helper-address on Cisco routers) is configured correctly.

Step 7: Test the connection again

After correcting the issue, run:

ipconfig /renew

ipconfig /all

Confirm that the device receives a valid IP address from the DHCP pool instead of 169.254.x.x.

Observation Table

Check	Possible finding	Solution
Network cable	Disconnected	Reconnect the cable
DHCP service	Disabled	Enable DHCP
DHCP pool	Address range exhausted	Expand the pool or free unused leases
Client configuration	Static or incorrect settings	Enable automatic IP configuration
DHCP relay	Missing or incorrect	Configure the correct relay address
VLAN configuration	Client on wrong VLAN	Correct VLAN assignment

5.Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

A. Common DNS Issues

No.	DNS Issue	Symptoms	Possible Causes	Recommended Solution
1	DNS server not responding	Websites do not open by domain name.	DNS server is down or unreachable.	Check DNS server connectivity and ensure the DNS service is running.
2	Incorrect DNS record	A domain resolves to the wrong IP address.	Incorrect A or CNAME record.	Verify and correct the DNS record.
3	DNS cache issue	A website resolves to an outdated IP address.	Old DNS information stored in the cache.	Run ipconfig /flushdns and retry the lookup.
4	Incorrect DNS configuration	Websites open by IP address but not by domain name.	Incorrect DNS server address on the client.	Check ipconfig /all and configure a valid DNS server.

B. Common DHCP Issues

No.	DHCP Issue	Symptoms	Possible Causes	Recommended Solution
1	DHCP server unavailable	Client cannot obtain an IP address and may receive an APIPA address (169.254.x.x).	DHCP service is stopped or the server is disconnected.	Check network connectivity and enable the DHCP service.
2	DHCP scope exhaustion	New devices cannot obtain an IP address.	All available addresses in the DHCP pool are leased.	Expand the DHCP pool or release unused leases.
3	Incorrect default gateway	Client accesses local devices but cannot reach external networks.	Incorrect gateway configured in the DHCP pool.	Correct the default gateway in the DHCP server settings and renew the lease.
4	DHCP relay misconfiguration	Clients on a different subnet fail to obtain IP addresses.	Missing or incorrect DHCP relay configuration.	Configure the correct ip helper-address on the router interface.

Commands Used for Troubleshooting

Command	Purpose
nslookup www.google.com	Checks DNS name resolution.
ipconfig /all	Displays IP, gateway, DHCP, and DNS configuration.
ipconfig /flushdns	Clears the local DNS resolver cache.
ipconfig /release	Releases the current DHCP IPv4 lease.
ipconfig /renew	Requests a new DHCP IPv4 lease.
ping 8.8.8.8	Tests IP connectivity to a public destination.